

SSR Polygonal Search Area

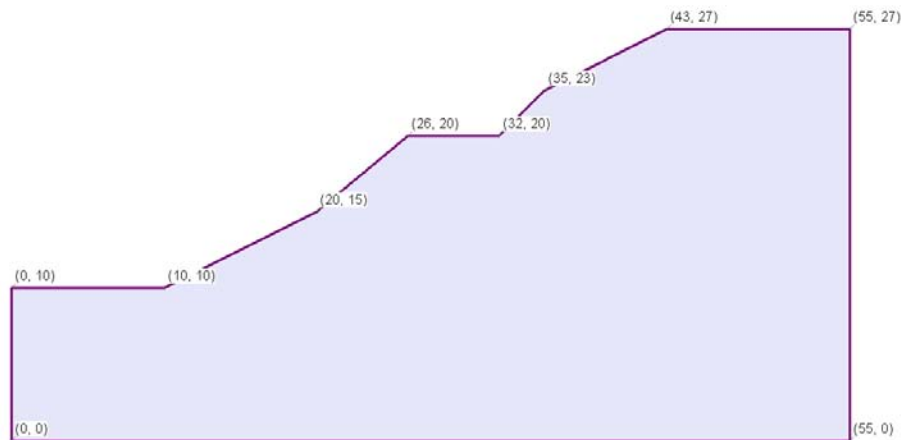
In this tutorial, *Phase2* is used to determine the factor of safety of a slope using the shear strength reduction (SSR) method. The SSR Polygon Search Area option is used to focus the SSR analysis to a specific region of the slope, in order to filter out local bench failures and determine a more important global failure mechanism.

The finished tutorial can be found in the **Tutorial 22 SSR Polygonal Search Area.fez** file. All tutorial files installed with *Phase2* 8.0 can be accessed by selecting File > Recent Folders > Tutorials Folder from the *Phase2* main menu.

Topics Covered

- Shear Strength Reduction
- SSR Polygonal Search Area

Geometry



Model

Start the *Phase2* Model program.

Project Settings



Open the **Project Settings** dialog from the **Analysis** menu and make sure the General tab is selected. Define the Units of Measurement as being “Metric, stress as kPa”.

Select the Strength Reduction tab. Turn on the Determine Strength Reduction factor checkbox. This enables the SSR analysis. Leave the various SSR settings at the default values. Close the Project Settings dialog by pressing the OK button.

Boundaries



This model only requires an External boundary to define the geometry. Select the **Add External** option in the **Boundaries** menu and enter the coordinates shown in the figure at the beginning of this tutorial.

Material Properties



Select **Define Materials** from the **Properties** menu. You will see the default properties for Material 1. Make sure the Initial Element Loading is set to Field Stress & Body Force (both in-situ stress and material self-weight are applied). Enter 20 kN/m³ for the Unit Weight. For Elastic Properties, make sure that Isotropic is the selected Elastic Type, and then enter 50000 kPa for the Young’s Modulus. For Strength Parameters, make sure the Failure Criterion is set to Mohr-Coulomb. Set the Material Type to Plastic, meaning the material can yield/fail. Set the peak and residual Tensile Strength to 0 kPa, the peak and residual Cohesion to 5, and the peak and residual Friction Angle to 30. Leave the dilation angle at 0. The dialog should look like this:

Define Material Properties

Material 1

Name: Material 1 Material Color: Material Color

Initial Element Loading: Field Stress & Body Force Unit Weight: (kN/m3): 20

Elastic Properties

Elastic Type: Isotropic Poisson's Ratio: 0.3

Young's Modulus (kPa): 50000 ☐ Young's Modulus (resid) (kPa): 20000

E1 (kPa): 20000 E2 (kPa): 20000 Ez (kPa): 20000

v12: 0.2 v13: 0.2 v23: 0.2

Strength Parameters

Failure Criterion: Mohr Coulomb Material Type: Plastic

Tensile Strength (peak) (kPa): 0 Dilation Angle (deg): 0

Fric. Angle (peak) (deg): 30 Fric. Angle (resid) (deg): 30

Cohesion (peak) (kPa): 5 Cohesion (resid) (kPa): 5

Tensile Strength (resid) (kPa): 0

☐ Stage Properties ☐ Datum Dependent

Define Factors... Define Properties...

Unsaturated Shear Strength

Phi b: 0 Air Entry (kPa): 0

Copy To... Statistics... ☐ Show only properties used in model OK Cancel

Press the OK button to save the properties and close the dialog.

Field Stress



Because the top of the model represents the true ground surface, we want to use a gravity field stress. Go to the **Loading** menu and select **Field Stress**. For Field Stress Type select Gravity and click the check box for "Use actual ground surface". Leave all other values as default.

Field Stress Properties

Field Stress Type: Gravity

☒ Use actual ground surface ☐ Use effective stress ratio ☐ Use variable stress ratio

Ground Surface Elevation (m): 0

Unit Weight of Overburden (kN/m3): 27

Total Stress Ratio (horiz/vert in plane): 1

Total Stress Ratio (horiz/vert out-of-plane): 1

Locked-in horizontal stress (in plane) (kPa, Comp. +): 0

Locked-in horizontal stress (out-of-plane) (kPa, Comp. +): 0

OK Cancel

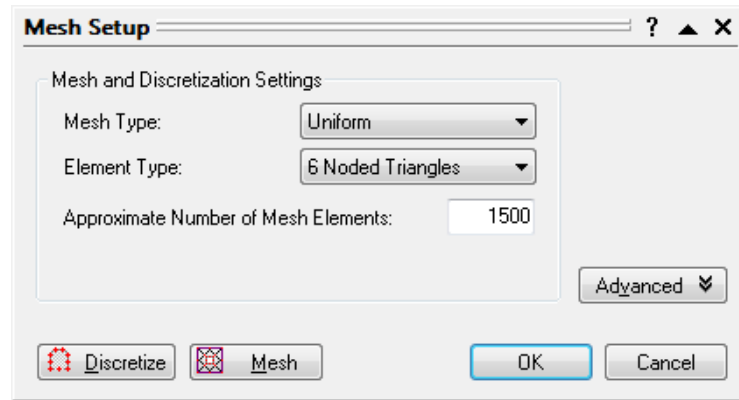
Statistics... Advanced >>

Click OK to close the dialog.

Mesh



Now generate the finite element mesh. Select the **Mesh Setup** option in the **Mesh** menu. Set the Mesh Type to Uniform. Leave the default number of elements (1500) but set the Element Type to 6 Noded Triangles. Click the Discretize button followed by the Mesh button.



Close the Mesh Setup dialog by selecting the OK button.

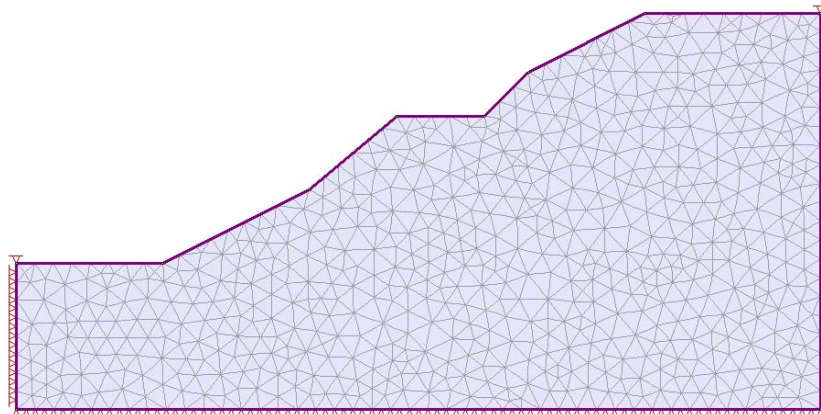
Boundary conditions



By default, all of the external boundary segments are fixed. Since the top of this model represents the actual ground surface, we need to free the top surface. Go to the **Displacements** menu and select **Free**. Click on the seven sections that make up the top boundary and hit Enter. You will now see that the fixed boundary conditions have disappeared from the top boundary.



We must, however, re-establish the fixed boundary condition for the upper left and upper right vertices of the slope. Select **Restrain X,Y** from the Displacement menu and click on the right and left vertical boundaries. Your model should now look like this:



Save the model using the **Save** option in the **File** menu.

Compute



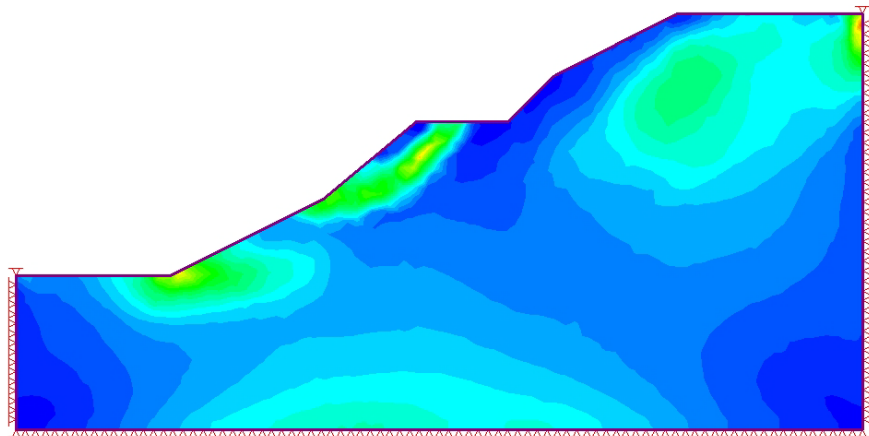
We will first analyze the model without specifying any SSR search region. Press Compute to perform the SSR analysis.



Once the model has finished computing (Compute dialog closes), select **Interpret** in the **Analysis** menu to view the results.

Interpret

The Interpret program starts and reads the results of the analysis. By default the maximum shear strain contours are displayed. The critical shear strength reduction factor (SRF) = 1.37.



If you click on the tabs for higher SRF factors, you will see the development of a clear failure surface, for the bench in the middle of the slope.

However, there may be other slope failures with very similar SRF values which a global search, or a rectangular SSR search window, will fail to find. We will now demonstrate how to use the polygonal SSR search window option to focus the SSR analysis on a specific region.

Model with polygonal search area



Go back to the *Phase2* model program by clicking on the Model button.

We now wish to perform an SSR analysis that excludes the small 'local' failure observed in the previous analysis. This can be done by specifying a polygonal SSR Search Area.

In general, an **SSR Search Area** is implemented in the SSR analysis as follows:

1. A material can only fail if the Material Type = Plastic in the Define Material Properties dialog. If a Material Type = Elastic then the material cannot "fail", and the material strength parameters are not applicable.
2. In general, the SSR analysis is only applicable for materials which have Material Type = Plastic.
3. When you define an SSR Search Area, what this does is effectively make the Material Type = Elastic for all finite elements which are **outside** of the SSR Search Area. NOTE:
 - This is done on an individual element basis. The original material properties are not actually changed. Instead, a new "elastic" version of each material is created, and assigned to each element which is outside of the SSR Search Area.
 - All finite elements **within OR crossing** an SSR Search area, are considered to be part of the SSR Search Area. Only elements which are entirely outside of a search area are given Elastic properties.
4. Therefore, failure can only occur within an SSR Search Area, during the SSR analysis, since all finite elements outside of the search area(s) are assumed to be Elastic.

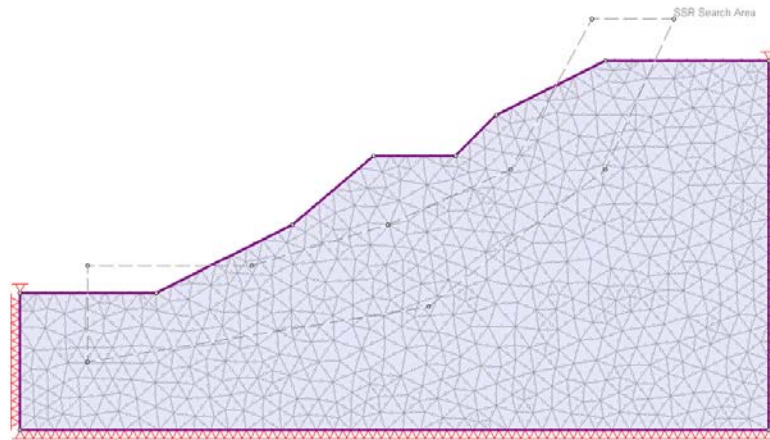
The use of a rectangular SSR search area is described in Tutorial 10. However, for this problem, we need to define a more complicated shape for the search area – hence a polygon.

Define SSR Polygon Search Area

In order to avoid the minimum SRF value bench failure, we need to define a polygon which misses this section of the slope. Go to the **Analysis** menu and select the **SSR Search Area > Define SSR Search Area (polygon)** option. Enter the following coordinates:

5, 5
 5, 12
 17, 12
 27, 15
 36, 19
 42, 30
 48, 30
 43, 19
 30, 9
 'c' to close the boundary.

The model will now look like this:



Save the file under a different name and then hit **Compute**.

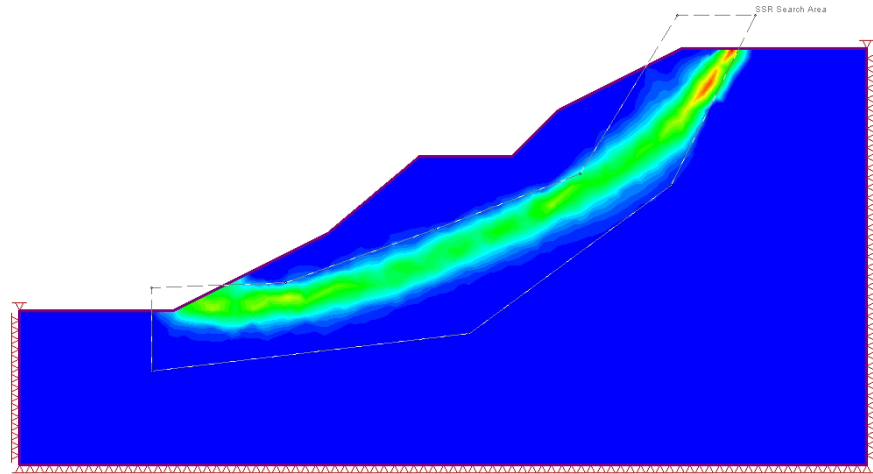


Once the model is done computing, hit the Interpret button to go back into the Interpreter.

Interpret

Examining the results, you can see the Critical SRF is now 1.41. Click through the higher SRF tabs. It is apparent that the SSR Search polygon has revealed a more important global failure mechanism which includes the entire slope, rather than just a single bench.

The figure below shows the Maximum shear strain at SRF = 1.5. Without the polygonal search area, we wouldn't be able to find this failure (i.e. a rectangular search window that included this large area would also include the bench in the middle of the slope, which has only a slightly lower critical SRF).



This concludes the SSR Polygonal Search Area tutorial.